RΓ(U,−) COMMUTES WITH R LIM; MILNOR SHORT EXACT SEQUENCES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- RF commutes with R lim. Injectives, Lemma injectives-lemma-RF-commutes-with-Rlim. RΓ(U,−) commutes with derived limits.
- Comparison of derived limits. More on Algebra, Remark more-algebra-remark-compare-derived-limit. Milnor short exact sequences for H^m of a derived limit.

Source reference. Original reference 08U1: sources/stacks-chow/tags.txt, source lines 11184–11184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 08U5: sources/stacks-chow/tags.txt, source lines 11188–11188. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. For $U \subset X$ open the functor $R\Gamma(U, -)$ commutes with $R\lim$. Moreover, there are short exact sequences*

$$0 \to R^1 \lim H^{m-1}(U, K_n) \to H^m(U, R\lim K_n) \to \lim H^m(U, K_n) \to 0$$

for any inverse system (K_n) in $D(\mathcal{O}_X)$ and any $m \in \mathbf{Z}$.

Proof. The first statement follows from Injectives, Lemma 08U1. Then we may apply More on Algebra, Remark 08U5 to $R\lim R\Gamma(U, K_n) = R\Gamma(U, R\lim K_n)$ to get the short exact sequences. □

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.